DEEP LEARNING • MEDICAL IMAGE CLASSIFICATION

Pneumonia Detection Using Chest X-Ray Images

An EfficientNetB0 transfer-learning system that classifies chest X-rays as NORMAL or PNEUMONIA.

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Reading X-Rays at Scale Is Slow and Subjective

1

What Pneumonia Is

An infection that inflames the air sacs in one or both lungs, visible on a chest X-ray as areas of opacity.

2

Why Early Detection Matters

Faster diagnosis leads to faster treatment, which is critical in severe or fast-progressing cases.

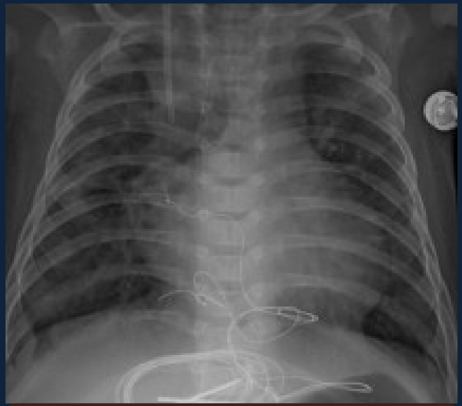
3

The Manual Bottleneck

Radiologists must visually review large volumes of X-rays one by one — a process that is time-consuming and dependent on individual expertise.

ACTUAL DATASET SAMPLES

NORMAL



PNEUMONIA

Subtle visual differences between healthy and infected lungs are exactly what a trained model learns to distinguish.

02 · THE PROJECT IDEA

From X-Ray Image to Instant Prediction

A single end-to-end pipeline turns a raw chest X-ray into a clear, automated classification — designed to support, not replace, clinical judgment.

1

Chest X-Ray

Raw input image

→

2

Image Processing

Resize · normalize · batch

→

3

Deep Learning Model

EfficientNetB0 classifier

→

4

Prediction

NORMAL or PNEUMONIA

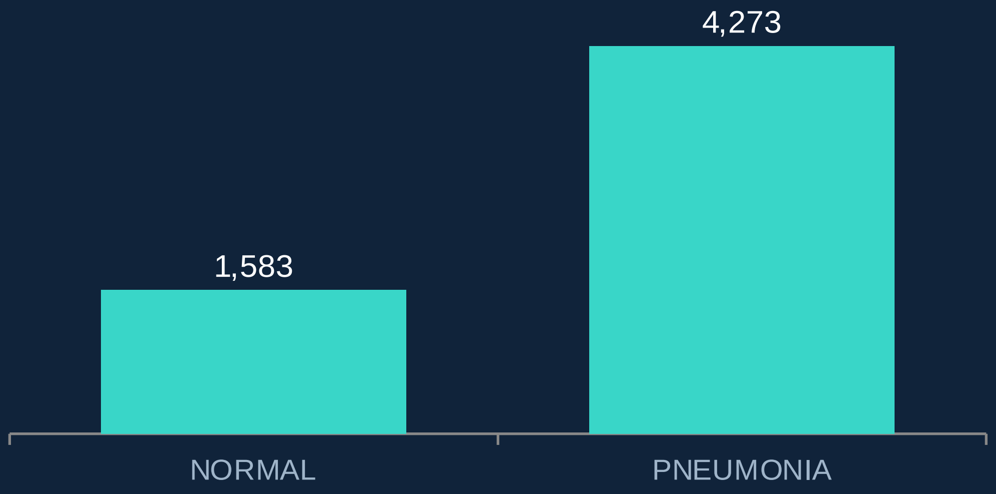
THE VALUE

A trained model can screen an X-ray in a fraction of the time it takes to review manually — offering a fast, consistent second opinion that scales to large volumes of images.

Chest X-Ray Pneumonia Dataset

Public Kaggle dataset (paultimothymooney/chest-xray-pneumonia) · Binary classes: NORMAL & PNEUMONIA

CLASS DISTRIBUTION — AFTER CLEANING & MERGING



5,856 total images · 1,583 NORMAL · 4,273 PNEUMONIA

Re-split into a fresh 70 / 15 / 15 train–validation–test partition (seed = 42):

4,099 Train

877 Validation

880 Test

DATASET AT A GLANCE

- **Original split**
  Train 5,216 · Val 16 · Test 624 images (pre-cleaning)
- **Cleaning result**
  0 invalid files and 0 corrupted images found
- **Image inspection**
  Sampled 500 images — 472 grayscale, 28 RGB
- **Native resolution**
  Width 400–2,572 px (avg 1,311) · Height 138–2,570 px (avg 962)
- **Target input size**
  Standardized to 224 × 224 for the model

A Clean Pipeline Before a Single Layer Trains

1

Data Cleaning

Removed non-image / invalid files

0 removed

2

Corrupted Image Detection

Fully loaded every file to catch truncated JPEGs

0 found

3

Image Inspection

Checked color mode & resolution on a 500-image sample

472 gray · 28 RGB

4

Resizing

Standardized every image to a fixed input size

224 × 224

5

Dataset Preparation

Merged, re-split, cached & prefetched with tf.data

70/15/15 split

WHY DATA QUALITY MATTERS

Medical images are sensitive to noise. Corrupted files, inconsistent formats, or mismatched sizes can silently degrade training — cleaning and standardizing the input is what makes the model's learning reliable.

Why EfficientNetB0 + Transfer Learning?

EfficientNetB0, in Plain Terms

A convolutional neural network pretrained on ImageNet that balances accuracy and computational cost through balanced (compound) scaling of depth, width, and resolution.

Transfer Learning

Instead of learning from scratch, the model starts from ImageNet-learned visual features — edges, textures, shapes — and adapts them to chest X-rays.

Why It Fits Medical Imaging

Labeled medical data is limited. Reusing a strong pretrained backbone lets the model reach good performance with a comparatively small, focused dataset.

PRETRAINED BACKBONE, NEW HEAD

EfficientNetB0 Backbone

ImageNet pretrained weights · initially frozen

FROZEN

↓

New Classification Head

GlobalAveragePooling → Dropout(0.3) → Dense + Softmax

TRAINABLE

06 · MODEL ARCHITECTURE

Input to Prediction, Layer by Layer

KEY DETAIL

The convolutional base is initially frozen — only the new pooling, dropout, and dense layers train in Stage 1. The base is selectively unfrozen later for fine-tuning.

Input Image — 224 × 224 × 3

↓

EfficientNetB0 — Pretrained convolutional backbone

↓

Global Average Pooling — Flattens feature maps

↓

Dropout (0.3) — Regularization

↓

Dense + Softmax — 2-class output layer

NORMAL | PNEUMONIA

Two-Stage Training Strategy

STAGE 1 · FROZEN BASE

EfficientNetB0 frozen · Adam optimizer · 10 epochs

Train accuracy: 85.6% → 94.7%
Val accuracy: 92.9% → 96.0%

(epoch 1 → epoch 9 of 10)

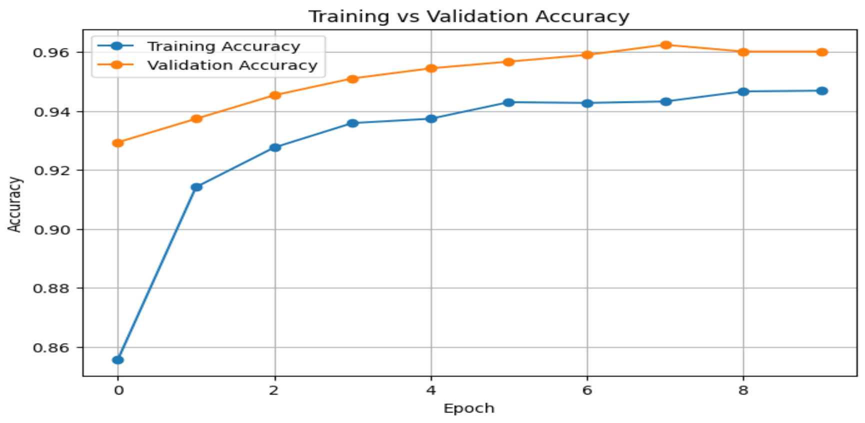
STAGE 2 · FINE-TUNING

Last 30 layers unfrozen · Adam @ lr = 1e-5 · 10 epochs

Train accuracy: 90.3% → 95.7%
Val accuracy: 95.8% → 96.7%

(epoch 1 → epoch 9 of 10)

Fine-tuning gently adapts the pretrained filters to the specific textures of chest X-rays, using a very small learning rate to avoid destroying the pretrained knowledge.



Training vs. Validation Accuracy — Stage 1 (actual run)

TEST-SET RESULT AFTER EACH STAGE

93.86% after Stage 1

94.66% after Stage 2 (final)

Test loss improved from 0.1551 → 0.1414

Final Performance on the Held-Out Test Set

94.66%
Final test accuracy (880 images)

0.1414
Final test loss

+0.80 pt
Accuracy gained via fine-tuning

NOTE

Precision, recall, F1 and a confusion matrix were not computed in this run — reported here only as an accuracy/loss result.

OVERALL PERFORMANCE vs. A SINGLE PREDICTION

The 94.66% figure above is an aggregate metric over the entire 880-image test set. The example below is one individual prediction — illustrative, not a measure of overall accuracy.

EXAMPLE PREDICTION FROM THE TEST SET

File: test/PNEUMONIA/train_person1469_bacteria_3824.jpeg

Predicted: PNEUMONIA

✓ Matches the ground-truth folder label

A Working Foundation for AI-Assisted Screening

WHAT THE PROJECT ACHIEVED

Built a complete pipeline — cleaning, re-splitting, and preprocessing a 5,856-image chest X-ray dataset, then training an EfficientNetB0 transfer-learning model that reached 94.66% test accuracy after two-stage fine-tuning.

POTENTIAL AS A SCREENING TOOL

As an assistive layer, a system like this could flag likely-abnormal X-rays for prioritized radiologist review — helping triage large volumes of images faster, without replacing clinical diagnosis.

FUTURE IMPROVEMENTS

- More robust evaluation: precision, recall, F1-score, confusion matrix
- Explainable AI (e.g. Grad-CAM) to visualize model attention
- Better generalization testing on external, independent datasets
- Deployment as a web application for interactive use

Pneumonia Detection Using Chest X-Ray Images — a transfer-learning foundation ready to grow into a validated clinical support tool.